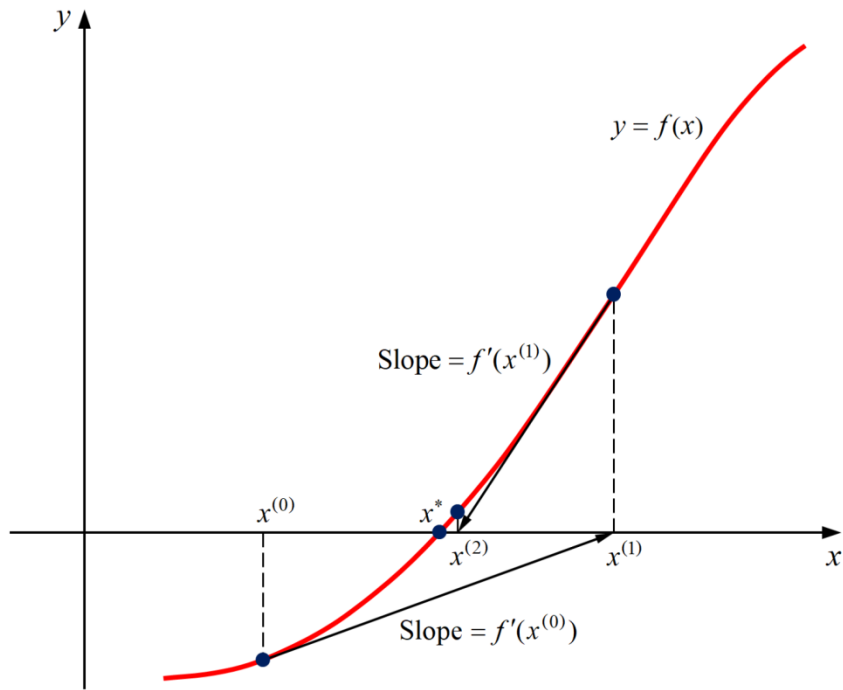


Newton-Raphson Method



Taylor Series Approximation:

$$f(x) \cong f(x_0) + f'(x_0) \cdot (x - x_0)$$

Rearranging:

$$f(x) = 0 \rightarrow x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Recurrence Relation:

$$x^{(k)} = x^{(k-1)} - \frac{f(x^{(k-1)})}{f'(x^{(k-1)})}$$

Relative Error:

$$\varepsilon^{(k)} = \frac{|x^{(k)} - x^{(k-1)}|}{|x^{(k)}|}$$

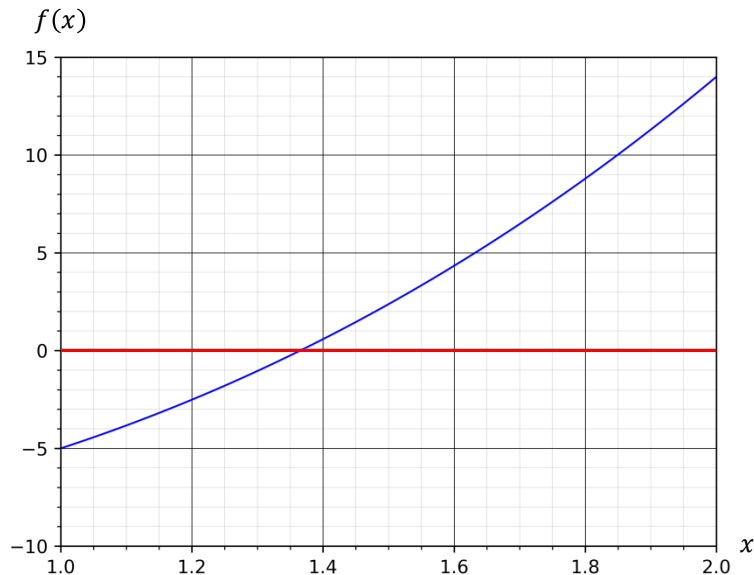
Relative Residual:

$$\varepsilon^{(k)} = \frac{|r^{(k)}|}{|r^{(0)}|} = \frac{|f(x^{(k)})|}{|f(x^{(0)})|}$$

Newton-Raphson Method

Example 02-04: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using Newton-Raphson method with initial guess $x^{(0)} = 1$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{f(x^{(k-1)})}{f'(x^{(k-1)})}; \quad \begin{aligned} f(x) &= x^3 + 4x^2 - 10 \\ f'(x) &= 3x^2 + 8x \end{aligned}$$



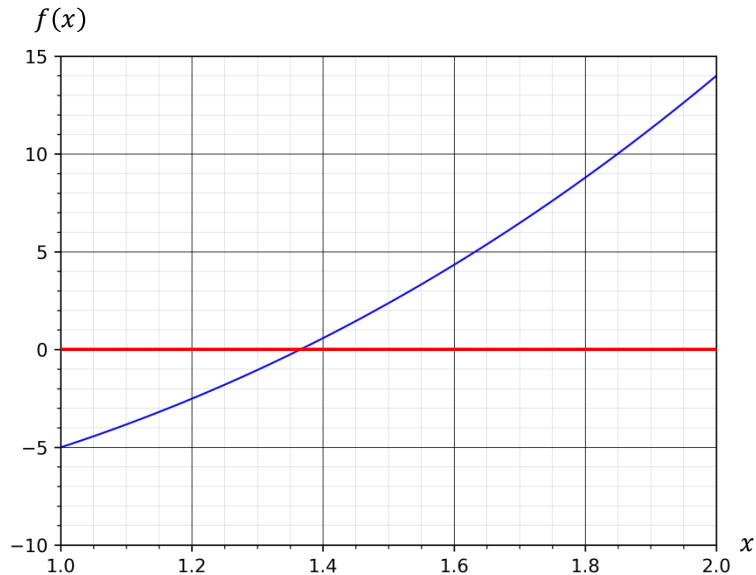
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$k = 1$

$$x^{(0)} = 1$$



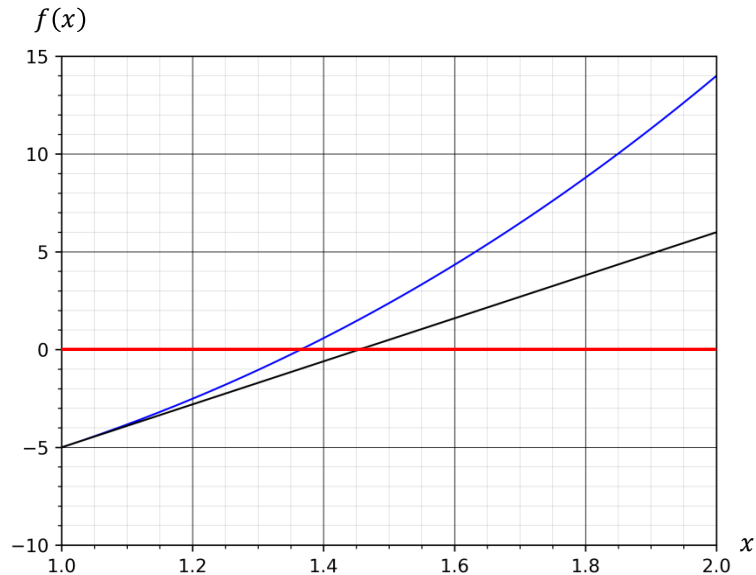
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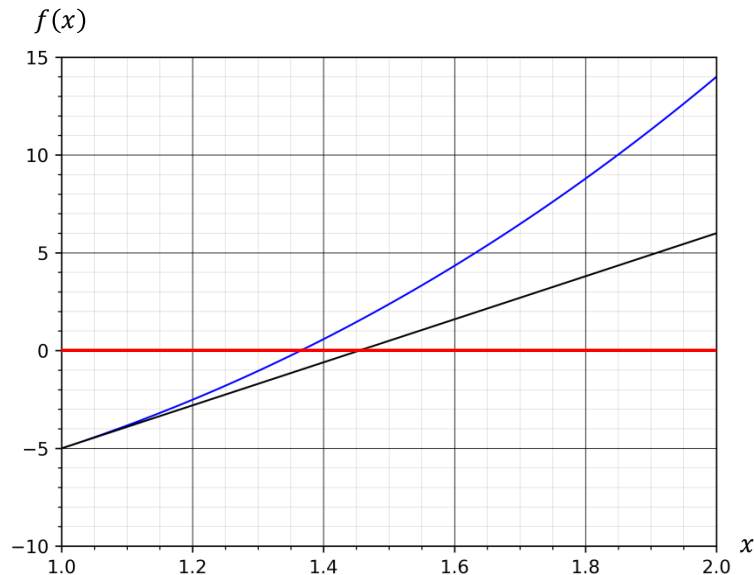
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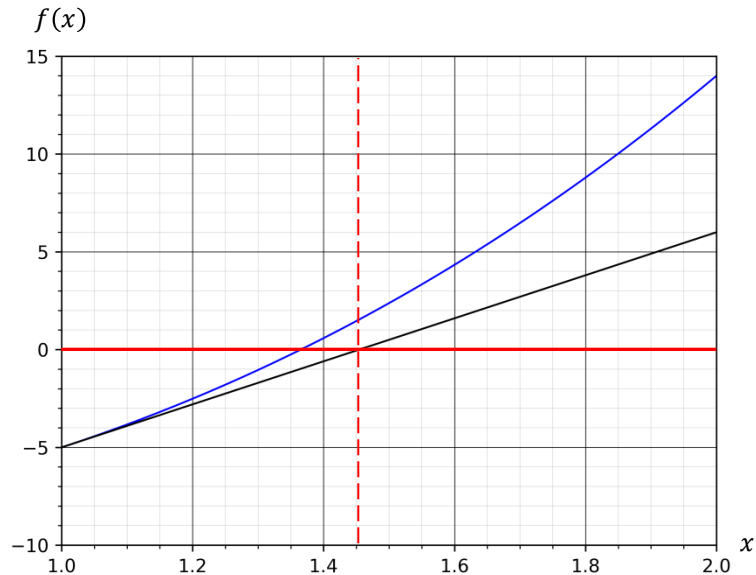
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Newton-Raphson Method

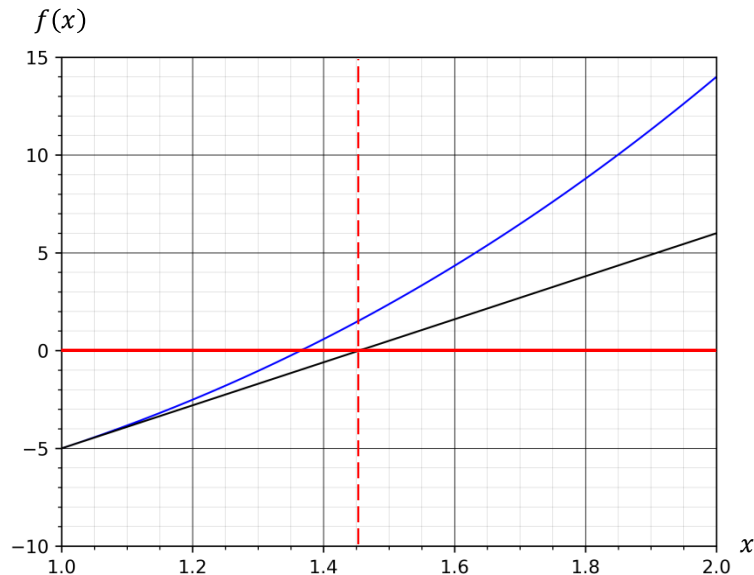
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Newton-Raphson Method

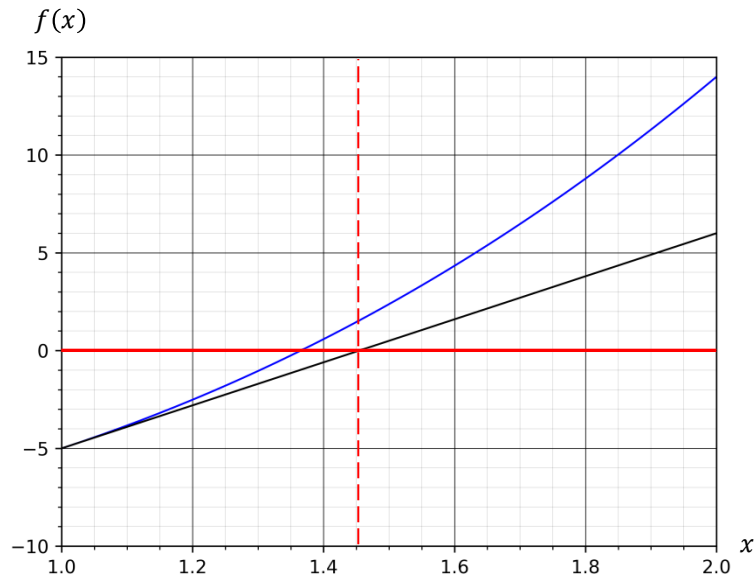
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Newton-Raphson Method

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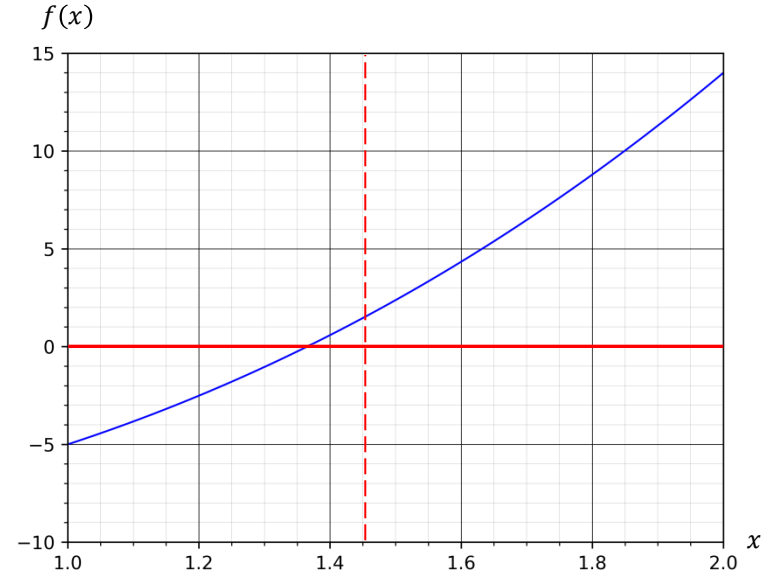
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$k = 2$

$$x^{(1)} = 1.454545$$



Newton-Raphson Method

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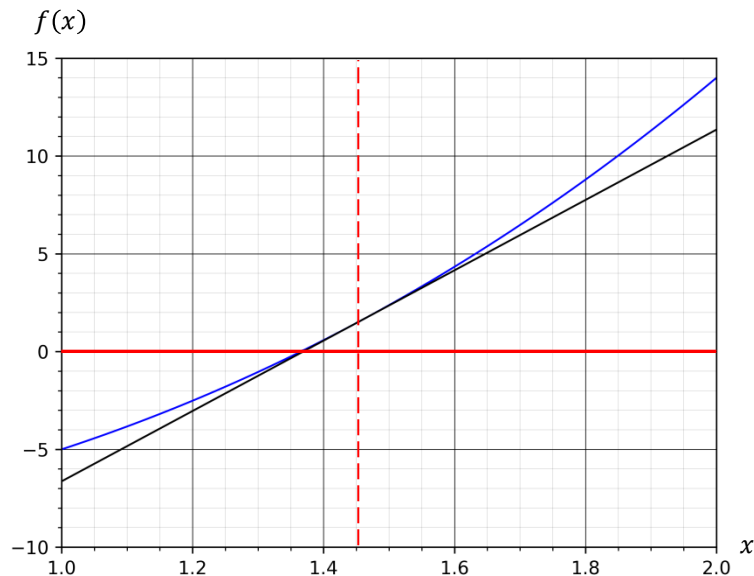
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Newton-Raphson Method

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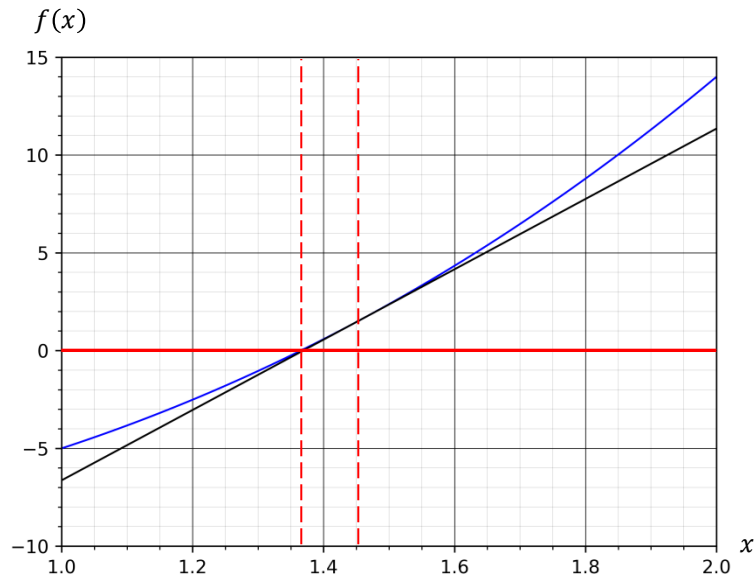
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$k = 2$

$$x^{(1)} = 1.454545 \rightarrow x^{(2)} = 1.368900$$



Newton-Raphson Method

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$k = 1$

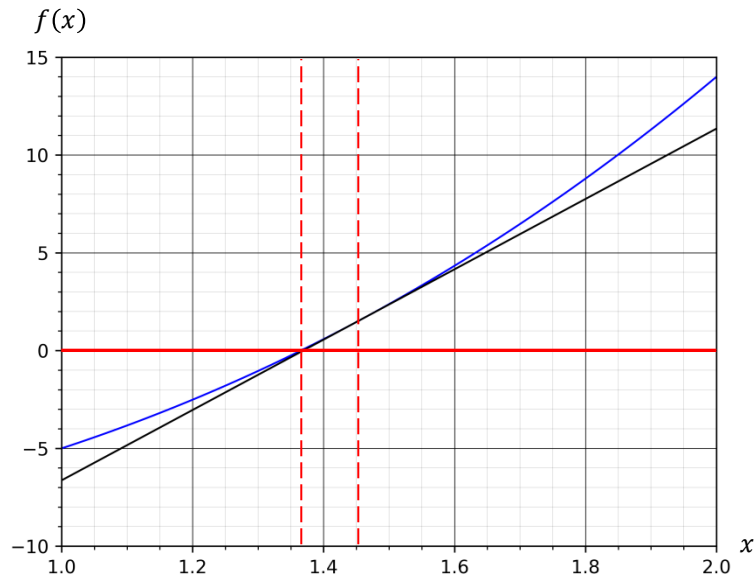
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$k = 2$

$$x^{(1)} = 1.454545 \rightarrow x^{(2)} = 1.368900$$

$$\varepsilon^{(2)} = \frac{|x^{(2)} - x^{(1)}|}{|x^{(2)}|} = \frac{|1.368900 - 1.454545|}{|1.368900|} = 0.062565 > \varepsilon_t$$



Newton-Raphson Method

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$k = 1$

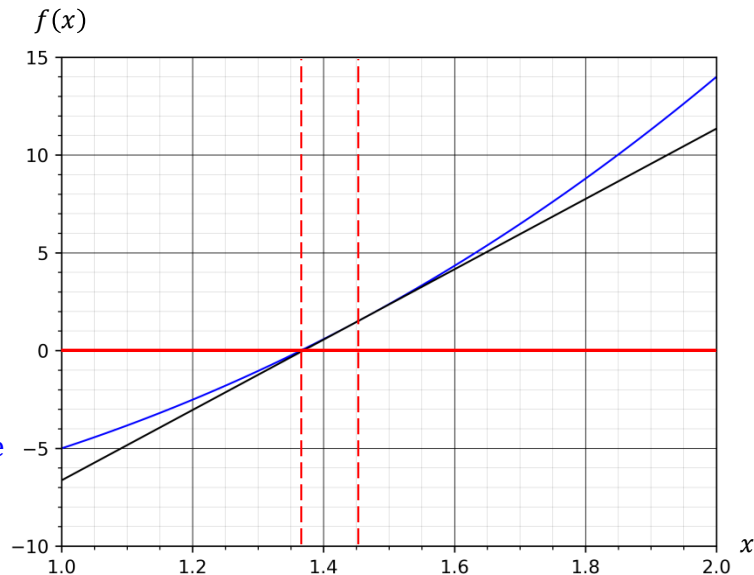
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Newton-Raphson Method

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$k = 3$

$$x^{(2)} = 1.368900$$

